

Gradient Descent: Worked Examples

David Puelz

The University of Austin

1 Example 1: A Nontrivial Two-Variable Function

1.1 The Function

Consider

$$f(x, y) = x^2 + 2xy + 2y^2 - 6x - 8y + 10.$$

This function has a *cross term* $2xy$, so it is not separable in x and y . The minimum does not lie at the “center” of uncoupled parabolas; we must use both partial derivatives together.

1.2 Computing the Gradient

Step 1. Partial derivative with respect to x (treat y as constant):

$$\frac{\partial f}{\partial x} = 2x + 2y - 6.$$

Step 2. Partial derivative with respect to y (treat x as constant):

$$\frac{\partial f}{\partial y} = 2x + 4y - 8.$$

Step 3. The gradient is

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) = (2x + 2y - 6, 2x + 4y - 8).$$

1.3 Finding the Critical Point Analytically

Set $\nabla f = (0, 0)$:

$$2x + 2y - 6 = 0 \quad \Rightarrow \quad x + y = 3$$

$$2x + 4y - 8 = 0 \quad \Rightarrow \quad x + 2y = 4$$

Subtracting the first from the second: $y = 1$. Then $x = 3 - y = 2$. So the critical point is

$$(x^*, y^*) = (2, 1).$$

We can verify $f(2, 1) = 4 + 4 + 2 - 12 - 8 + 10 = 0$ (the minimum value).

1.4 Finding the Critical Point via Gradient Descent

When we cannot solve $\nabla f = \mathbf{0}$ analytically, we use **gradient descent**:

$$(x^{(k+1)}, y^{(k+1)}) = (x^{(k)}, y^{(k)}) - \alpha \nabla f(x^{(k)}, y^{(k)}).$$

Iteration 0. Start at $(x^{(0)}, y^{(0)}) = (0, 0)$. Choose $\alpha = 0.2$.

$$\nabla f(0, 0) = (2(0) + 2(0) - 6, 2(0) + 4(0) - 8) = (-6, -8).$$

Iteration 1.

$$(x^{(1)}, y^{(1)}) = (0, 0) - 0.2(-6, -8) = (0 + 1.2, 0 + 1.6) = (1.2, 1.6).$$

Iteration 2.

$$\nabla f(1.2, 1.6) = (2.4 + 3.2 - 6, 2.4 + 6.4 - 8) = (-0.4, 0.8).$$

$$(x^{(2)}, y^{(2)}) = (1.2, 1.6) - 0.2(-0.4, 0.8) = (1.2 + 0.08, 1.6 - 0.16) = (1.28, 1.44).$$

Further iterations. Repeating this process, the sequence converges to $(2, 1)$. With $\alpha = 0.2$, after about 15–20 iterations we are very close to the true minimum.

Interpretation. At each step, we move in the direction of $-\nabla f$ (steepest descent). The gradient at $(0, 0)$ points in the direction $(-6, -8)$, i.e., toward larger x and y (since we subtract the gradient, we move opposite to it—toward the minimum).

2 Example 2: Least Squares with One Predictor

2.1 The Model

We have data $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$. The model is

$$\hat{y}_i = \beta_0 + \beta_1 x_i,$$

where β_0 is the intercept and β_1 is the slope. We minimize the **sum of squared errors** (SSE):

$$L(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2.$$

2.2 Computing the Gradients (Step-by-Step)

Step 1. Write the loss in expanded form. For each term i :

$$(y_i - \beta_0 - \beta_1 x_i)^2.$$

Let $e_i = y_i - \beta_0 - \beta_1 x_i$ (the residual for observation i). Then $L = \sum_i e_i^2$.

Step 2. Derivative with respect to β_0 .

Using the chain rule: $\frac{\partial}{\partial \beta_0}(e_i^2) = 2e_i \cdot \frac{\partial e_i}{\partial \beta_0}$. Since $e_i = y_i - \beta_0 - \beta_1 x_i$,

we have $\frac{\partial e_i}{\partial \beta_0} = -1$. So

$$\frac{\partial L}{\partial \beta_0} = \sum_{i=1}^n 2(y_i - \beta_0 - \beta_1 x_i)(-1) = -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i).$$

Step 3. Derivative with respect to β_1 .

Similarly, $\frac{\partial e_i}{\partial \beta_1} = -x_i$. So

$$\frac{\partial L}{\partial \beta_1} = \sum_{i=1}^n 2(y_i - \beta_0 - \beta_1 x_i)(-x_i) = -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) x_i.$$

Step 4. The gradient is

$$\nabla L = \left(\frac{\partial L}{\partial \beta_0}, \frac{\partial L}{\partial \beta_1} \right) = \left(-2 \sum_i (y_i - \beta_0 - \beta_1 x_i), -2 \sum_i (y_i - \beta_0 - \beta_1 x_i) x_i \right).$$

Step 5. Setting $\nabla L = (0, 0)$ and dividing by -2 gives the **normal equations**:

$$\begin{aligned} \sum_i (y_i - \beta_0 - \beta_1 x_i) &= 0 \\ \sum_i (y_i - \beta_0 - \beta_1 x_i) x_i &= 0 \end{aligned}$$

These have a closed-form solution (the familiar OLS formulas). But we can also find (β_0, β_1) via gradient descent when the closed form is inconvenient (e.g., many predictors, regularized regression).

2.3 Gradient Descent for Least Squares

Update rule:

$$(\beta_0^{(k+1)}, \beta_1^{(k+1)}) = (\beta_0^{(k)}, \beta_1^{(k)}) - \alpha \nabla L(\beta_0^{(k)}, \beta_1^{(k)}).$$

Numerical example. Suppose we have three points: $(1, 2)$, $(2, 3)$, $(3, 5)$.

Start at $(\beta_0, \beta_1) = (0, 0)$ with $\alpha = 0.1$.

Iteration 0. Residuals: $e_1 = 2 - 0 - 0 = 2$, $e_2 = 3$, $e_3 = 5$.

$$\frac{\partial L}{\partial \beta_0} = -2(2 + 3 + 5) = -20, \quad \frac{\partial L}{\partial \beta_1} = -2(2 \cdot 1 + 3 \cdot 2 + 5 \cdot 3) = -2(2 + 6 + 15) = -46.$$

$$(\beta_0^{(1)}, \beta_1^{(1)}) = (0, 0) - 0.1(-20, -46) = (2, 4.6).$$

Further iterations. Repeating, the sequence converges to the OLS solution. For this small dataset, the closed-form solution is $\beta_1 = \frac{\text{Cov}(x,y)}{\text{Var}(x)}$ and $\beta_0 = \bar{y} - \beta_1 \bar{x}$. Gradient descent recovers the same values.

3 Summary

- **Gradient:** $\nabla f = (\partial f / \partial x, \partial f / \partial y)$ points in the direction of steepest ascent.
- **Gradient descent:** $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \alpha \nabla f(\mathbf{x}^{(k)})$ moves toward a minimum.
- **Least squares:** The loss $L = \sum (y_i - \beta_0 - \beta_1 x_i)^2$ has gradients that lead to the normal equations; gradient descent is an iterative alternative that generalizes to complex models.